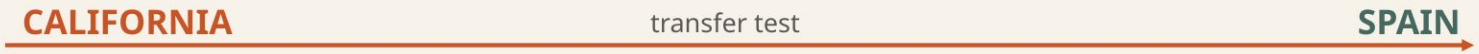
Can a wildfire model travel from California to Spain?

Examining wildfire-model generalizability through cross-region transfer

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CCIR Symposium



Wildfire danger is becoming a bigger-risk problem

"More dangerous" means larger burned area, longer seasons, higher exposure, and harder-to-predict events.

1.08M

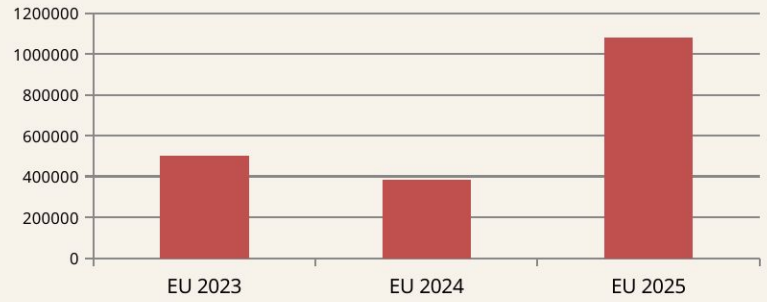
hectares burned in EU27 in 2025 — highest since EFFIS records began

1.08M

acres burned in California in 2024 across 8,110 wildfires

~500k

hectares burned in the EU in 2023 — fourth largest on record



Sources: [1] JRC / EFFIS 2025; [2] European Civil Protection 2025; [3] CAL FIRE 2024 incident archive; [4] Copernicus ESOTC 2023.

Risk rises when weather, fuel, and people line up

Avoid saying one fire was “caused by climate change”; say warming shifts the conditions that make fires more likely and severe.

Fire weather

Heat, low humidity, wind, and drought make fuels easier to ignite and spread.

Available fuel

Vegetation type, live fuel moisture, land cover, and fuel continuity shape burn potential.

Ignitions & exposure

Human-caused ignitions, suppression policy, land management, and settlement patterns change outcomes.

Transfer failure is scientifically useful: it may show where climate is not the whole story.

Sources: [1], [2], [4], [5].

The cost of being wrong is not abstract

Prediction matters because missed or poorly ranked risk turns into ecological, financial, and community harm.

1. Biodiversity loss and habitat fragmentation after large burns.

2. Smoke exposure, evacuations, fatalities, and community disruption.

3. Agricultural losses, infrastructure damage, insurance pressure, and public firefighting cost.

4. Planning failure: agencies cannot pre-position resources if risk is ranked incorrectly.

What better prediction changes

Earlier warning — More time to evacuate and stage crews.

Smarter allocation — Rank scarce firefighting resources before ignition risk spikes.

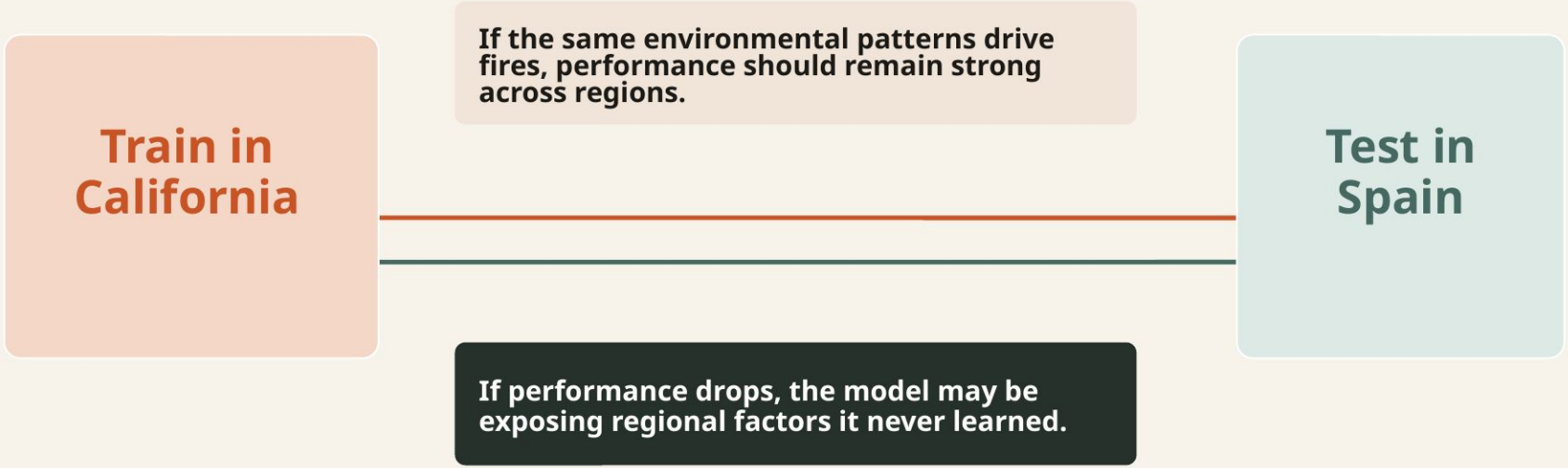
Targeted prevention — Focus fuel work and outreach where model risk is highest.

Better transfer tests — Know when a model is learning local quirks, not general fire physics.

Sources: [4], [5].

The question: does learned fire risk transfer?

California and Spain share fire-prone seasonal conditions, but geography, policy, land use, and ignition patterns differ.



The study tests generalization, not just accuracy

The important test is chronological and geographic: train on earlier years, then test on unseen time and unseen region.



Chronological split prevents leakage from nearby dates and makes the test more realistic.

Project code: wildfire_modelv2.py uses 42 variables and a train/validation/test temporal split.

The dataset turns daily fire conditions into 42 features

Each row represents a region–date–grid-cell event labeled as fire or non-fire.

Region	Rows	Fires	Fire share	Train	Validation	Test
California	107,612	26,903	25.0%	81,694	9,858	9,487
Spain	58,936	14,734	25.0%	47,596	5,755	5,585

Feature groups

Same-day weather	temperature, humidity, wind speed, precipitation
Rolling windows	30-, 90-, and 180-day precipitation and temperature averages
Vegetation & land cover	LAI high vegetation, LAI low vegetation, total LAI, land-cover class
Lagged weather	T−1 through T−7 temperature, precipitation, humidity, and wind

Sources: Project logs and scripts; feature categories based on ERA5, MODIS, LAI, NLCD/CORINE inputs.

Tree models are the right level of complexity here

The goal is explainable cross-region transfer, not a black-box leaderboard.

Random Forest

Many decision trees vote together. It is robust, interpretable, and a strong baseline for tabular environmental data.

XGBoost

Sequential trees correct previous errors. It often performs better on structured tabular data and handles nonlinear interactions.

Why not neural networks?

The dataset is tabular and modest by deep-learning standards; explainability and transfer diagnostics matter more than model complexity.

Model choices were matched to the research question

1. Good for mixed weather + land-cover inputs
2. Can rank feature importance
3. Can be evaluated with SHAP
4. Avoids overclaiming with limited data

Methods reference: [5] McNorton et al. used RF and XGBoost for global Probability-of-Fire forecasting.

A 75% accurate model can still miss every fire

Both regional datasets contain 25% fire cases and 75% non-fire cases—a 3:1 class imbalance.

Why accuracy can mislead



Naive model: predict "No Fire" for every case

		Predicted: No Fire	Predicted: Fire
Actual	No Fire	TN = 750	FP = 0
	Fire	FN = 250	TP = 0

Accuracy = (750 + 0) / 1,000 = 75%

Recall = 0 / (0 + 250) = 0% • **F1 = 0**

How the metrics are calculated

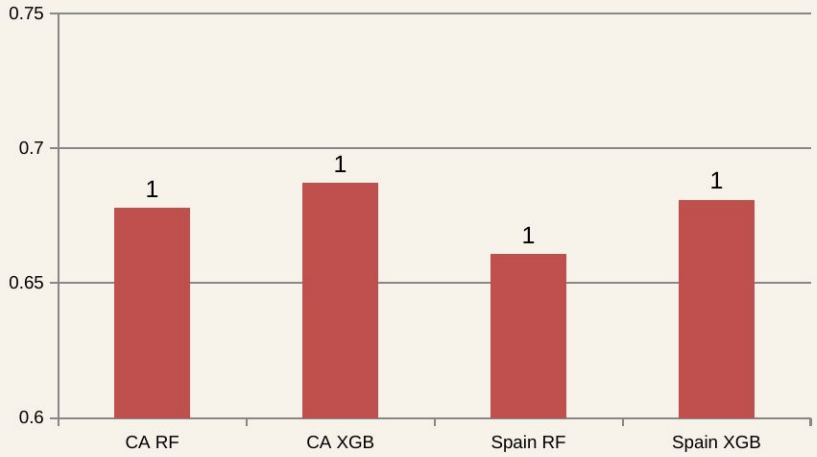
Accuracy	**(TP + TN) / (TP + TN + FP + FN)**	Overall correctness. Easily inflated when non-fire cases dominate.
Precision	**TP / (TP + FP)**	When the model predicts fire, how often is it right?
Recall	**TP / (TP + FN)**	Of all actual fires, how many did the model catch?
F1 score	**2 × Precision × Recall / (Precision + Recall)**	A single score that penalizes weak precision or weak recall.
ROC-AUC	**P(score_fire > score_non-fire)**	Across thresholds, how well does the model rank fires above non-fires?

Accuracy asks "How often was the model right?" Recall asks the more important question here: "How many fires did it miss?"

Within-region models work: XGBoost edges Random Forest

Both regions achieve strong AUC near 0.90; XGBoost is slightly stronger on F1 and recall.

Model	Accuracy	Precision	Recall	F1	AUC
CA RF	0.822	0.592	0.794	0.678	0.901
CA XGB	0.826	0.597	0.809	0.687	0.904
Spain RF	0.817	0.593	0.747	0.661	0.884
Spain XGB	0.824	0.602	0.784	0.681	0.903

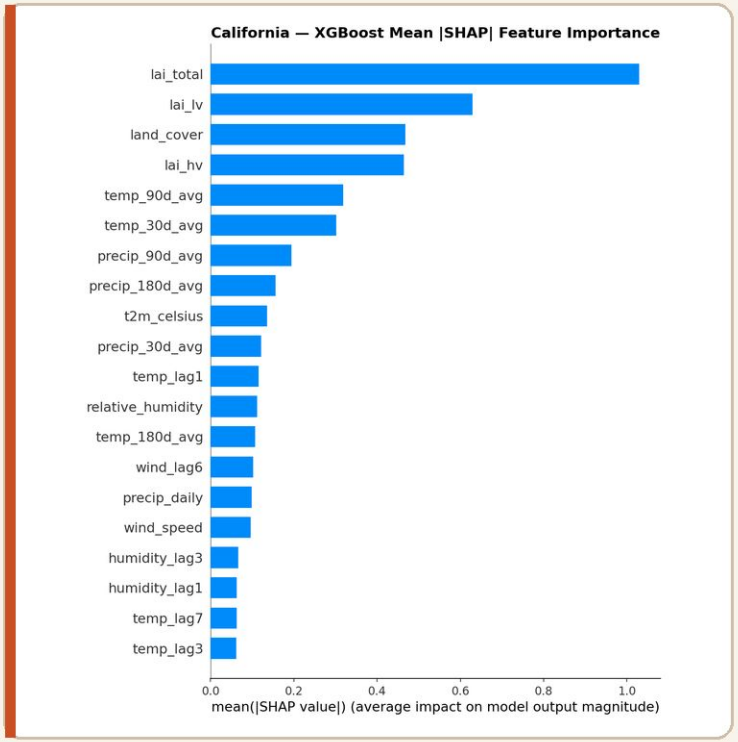


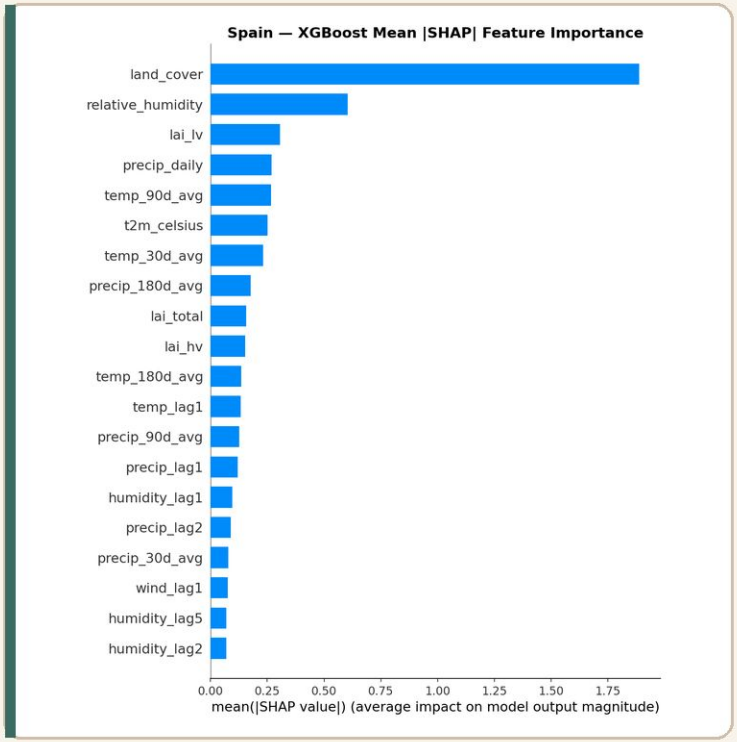
Takeaway: the models learn meaningful regional fire-risk patterns before transfer is attempted.

Project model output: RF and XGBoost test metrics from wildfire_modelv2.py / user run log.

Vegetation dominates California; land cover dominates Spain

Mean |SHAP| ranks average impact magnitude. It shows importance, not whether a feature raises or lowers predicted fire risk.

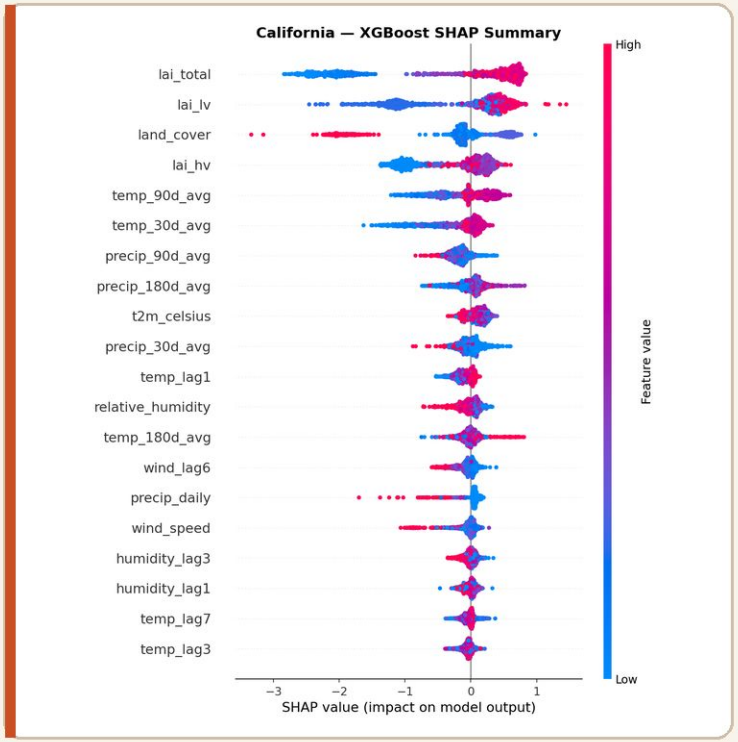


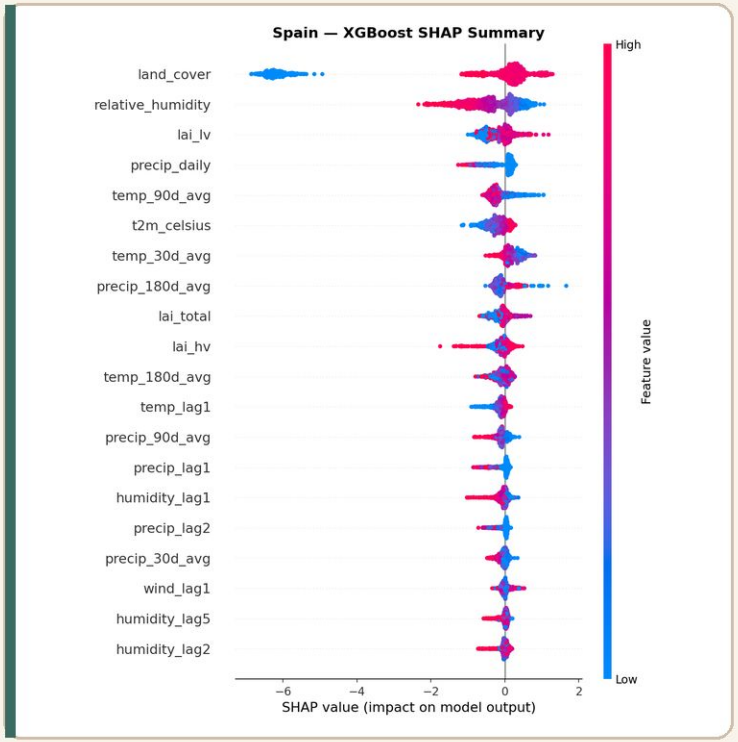


The hierarchy of wildfire drivers changes by region—an early warning that the models may not transfer cleanly.

The same feature can push risk differently across regions

Each dot is one observation. Red indicates a high feature value; blue indicates a low value. Positive SHAP values push the model toward predicting fire.





Direction, spread, and threshold behavior differ across regions—not merely the feature ranking.

The model works locally, then breaks across regions

That is exactly why cross-region testing matters.

1. Within-region performance is strong enough to take the model seriously.

2. Transfer performance drops, especially Spain → California.

3. The drop suggests wildfire risk is not only weather + vegetation; regional human and policy factors likely matter.

4. Counterfactuals are the next phase: test how risk changes when climate variables are systematically shifted.

Main message: Generalization failure is not failure of the project — it is the result.